

On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

April 2026

Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \dots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$\log k(n) = (\log n)^{1/2+o(1)},$$

confirming the proposed estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ and showing it is sharp. The lower bound follows from a counting/pigeonhole argument on smooth numbers. The upper bound uses the Hildebrand–Tenenbaum equidistribution theorem for smooth numbers in short intervals.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. For positive integers k and n , a *k-run below n* is an interval $\{m+1, \dots, m+k\}$ with $m \leq n$ such that $P^+(m+j) > k$ for every $1 \leq j \leq k$. Equivalently, none of $m+1, \dots, m+k$ is k -smooth. Define $k(n)$ to be the maximum k for which a k -run below n exists.

Remark 3. A k -run below n is exactly a gap of length $\geq k$ in the set of k -smooth integers inside $[1, n+k]$: an interval $[m+1, m+k]$ with $m \leq n$ that contains no k -smooth integer.

2 Smooth Number Background

Theorem 4 (Dickman [1], de Bruijn [2]). *Let $\rho: [0, \infty) \rightarrow (0, 1]$ be the Dickman function, defined by $\rho(u) = 1$ for $u \in [0, 1]$ and $u\rho'(u) = -\rho(u-1)$ for $u > 1$. For $u > 0$ fixed and $y = x^{1/u}$,*

$$\Psi(x, y) = x \rho(u) (1 + o(1)) \quad \text{as } x \rightarrow \infty.$$

The same holds uniformly for $1 \leq u \leq (\log x)^{3/5-\varepsilon}$ (Hildebrand–Tenenbaum [3]).

Lemma 5 (Dickman bound). *For all $u \geq 1$, $\rho(u) \leq u^{-u}$.*

Proof. One checks inductively that $\rho(u) \leq 1/(u \Gamma(u)) \leq u^{-u}$; see [4, Ch. III.5]. $\square$

The short-interval equidistribution we need is the following.

Theorem 6 (Hildebrand–Tenenbaum [3], short intervals). *Fix $\varepsilon > 0$. Uniformly for $x \geq 2$, $\exp((\log \log x)^{5/3+\varepsilon}) \leq y \leq x$, and $u = \log x / \log y$,*

$$\Psi(x+y, y) - \Psi(x, y) = y \rho(u) (1 + o(1))$$

as $x \rightarrow \infty$.

3 The Lower Bound

Theorem 7 (Lower bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \geq \exp((\log n)^{1/2-\varepsilon}).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$ and $y = k_0$.

Step 1: The smooth numbers are sparse. Write $u_0 = \log n / \log k_0 = (\log n)^{1/2+\varepsilon} (1 + o(1))$. By Lemma 5,

$$u_0 \log u_0 = (1/2 + \varepsilon)(\log n)^{1/2+\varepsilon} \log \log n (1 + o(1)) \geq (\log n)^{1/2+\varepsilon/2}$$

for large n , so $\rho(u_0) \leq e^{-(\log n)^{1/2+\varepsilon/2}}$. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$, we have $e^{-(\log n)^{1/2+\varepsilon/2}} \ll 1/k_0$. By Theorem 4,

$$\Psi(n, k_0) = n \rho(u_0) (1 + o(1)) \leq \frac{n}{k_0}$$

for all sufficiently large n .

Step 2: Pigeonhole gives an empty block. Partition $\{1, 2, \dots, \lfloor n/k_0 \rfloor \cdot k_0\}$ into $B = \lfloor n/k_0 \rfloor$ consecutive blocks $I_i = [(i-1)k_0 + 1, ik_0]$ of length k_0 each. There are $B \approx n/k_0$ blocks and at most $\Psi(n, k_0) \leq n/k_0 \approx B$ smooth integers spread across them. In fact we showed $\Psi(n, k_0) \ll n/k_0$, so strictly fewer smooth integers than blocks. By pigeonhole, at least one block I_i contains no k_0 -smooth integer.

Step 3: The empty block is a k_0 -run. If $I_i = [(i-1)k_0 + 1, ik_0]$ contains no k_0 -smooth integer, then $P^+(j) > k_0$ for every $j \in I_i$. Setting $m = (i-1)k_0 \leq n$, the integers $m+1, \dots, m+k_0$ all have largest prime factor $> k_0$. This is a k_0 -run below n , so $k(n) \geq k_0$. $\square$

Remark 8. The CRT approach—choose distinct primes $q_1, \dots, q_k > k$ and solve $q_j \mid m+j$ by CRT—gives modulus $Q = \prod q_j$ with $\log Q \approx k \log k$. For $k = e^{(\log n)^{1/2-\varepsilon}}$ this gives $\log Q \gg \log n$, so $Q \gg n$ and $m < Q$ need not satisfy $m \leq n$. The CRT method yields only $k(n) \geq (1-o(1)) \log n / \log \log n$ (from $k \log k \leq \log n$), which is far weaker.

4 The Upper Bound

Theorem 9 (Upper bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \leq \exp((\log n)^{1/2+\varepsilon}).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer, so no k_1 -run below n exists.

Step 1: Hildebrand–Tenenbaum applies. We need $k_1 \geq \exp((\log \log(m+k_1))^{5/3+\varepsilon'})$ for some $\varepsilon' > 0$. Since $m \leq n$, we have $\log \log(m+k_1) = O(\log \log n)$, and $\log k_1 = (\log n)^{1/2+\varepsilon} \gg (\log \log n)^{5/3+\varepsilon'}$ for any $\varepsilon' > 0$ and large n . So Theorem 6 applies with $x = m$ and $y = k_1$ for every $m \in [n^{1/2}, n]$.

Step 2: The expected smooth count in $[m+1, m+k_1]$ is large. For $m \in [n^{1/2}, n]$, set $u_1 = \log m / \log k_1 \in [(\log n)^{1/2-\varepsilon}/2, (\log n)^{1/2-\varepsilon}]$. Theorem 6 gives

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) = k_1 \rho(u_1) (1 + o(1)).$$

The product $k_1 \rho(u_1)$ satisfies

$$\log(k_1 \rho(u_1)) = (\log n)^{1/2+\varepsilon} - u_1 \log u_1 (1 + o(1)).$$

Since $u_1 \leq (\log n)^{1/2-\varepsilon}$,

$$u_1 \log u_1 \leq (\log n)^{1/2-\varepsilon} \cdot (1/2 - \varepsilon) \log \log n = o((\log n)^{1/2+\varepsilon}).$$

Hence $\log(k_1 \rho(u_1)) = (\log n)^{1/2+\varepsilon}(1 - o(1)) \rightarrow +\infty$, so $k_1 \rho(u_1) \rightarrow \infty$ and the interval $[m+1, m+k_1]$ contains $k_1 \rho(u_1)(1 + o(1)) \geq 1$ smooth integers for large n .

Step 3: Small m . For $m < n^{1/2} \ll k_1$, every integer $j \leq k_1$ is k_1 -smooth (its prime factors are all $\leq j \leq k_1$). In particular $[m+1, m+k_1] \supseteq \{1, \dots, k_1\}$ for $m = 0$, and for general $m < k_1$ the interval contains integers $\leq k_1$ which are k_1 -smooth.

Conclusion. Every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer. Hence no k_1 -run exists below n , and $k(n) < k_1 = e^{(\log n)^{1/2+\varepsilon}(1+o(1))}$. $\square$

5 Main Result and Sharpness

Theorem 10 (Main Theorem). *As $n \rightarrow \infty$,*

$$\log k(n) = (\log n)^{1/2+o(1)}.$$

*The proposed bound $\log k(n) \leq (\log n)^{1/2+o(1)}$ is **true and sharp**.*

Proof. Theorems 7 and 9. $\square$

Proposition 11 (Precise asymptotics). $\log k(n) \sim \left(\frac{1}{2} \log n \cdot \log \log n\right)^{1/2}$ as $n \rightarrow \infty$.

Sketch. The threshold value of k is determined by the self-consistency equation $\rho(u) \asymp 1/k$ with $u = \log n / \log k$. Using $\rho(u) = e^{-u \log u (1+o(1))}$, writing $L = \log k$ and $\ell = \log n$, this becomes $u \log u \sim L$, i.e. $(\ell/L) \log(\ell/L) \sim L$, giving $L^2 \sim \ell \log(\ell/L)$. Setting $L = c\sqrt{\ell \log \ell}$ and expanding: $c^2 \ell \log \ell \sim \ell \cdot \frac{1}{2} \log \ell$, so $c = 1/\sqrt{2}$. Hence $\log k(n) \sim \left(\frac{1}{2} \log n \cdot \log \log n\right)^{1/2}$. $\square$

Remark 12 (Primitive sets). Each k -run $\{m+1, \dots, m+k\}$ is a primitive set: if $a \mid b$ with $a < b$ both in the run, then $b \geq 2a > m+k$, a contradiction. The condition $P^+(a) > k$ gives each element a prime certificate $> k$, making these runs a special class of primitive sets.

References

- [1] K. Dickman, *On the frequency of numbers containing prime factors of a certain relative magnitude*, Ark. Mat. Astr. Fys. **22A** (1930), 1–14.
- [2] N. G. de Bruijn, *On the number of positive integers $\leq x$ free of prime factors $> y$* , Indag. Math. **13** (1951), 50–60.
- [3] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.
- [4] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 3rd ed., 2015.